

Large language models can encode information about toxic or undesirable language within their internal representations, but this information is not always directly accessible or easily interpretable. Conventional toxicity classifiers typically operate on text directly, which can require additional models or extensive fine-tuning.

This project investigates whether toxicity can be detected from the hidden representations of a frozen language model. Using google/gemma-2-2b, we extract and mean-pool hidden states from text inputs and train a lightweight linear probe to distinguish between safe and toxic content.

The problem is therefore to determine whether internal representations of a language model contain sufficient information to reliably identify toxic inputs, and at which layer this information can be most effectively detected using a simple linear classifier.